AI-Driven Early Detection of Thyroid Cancer in Rural Nepal

Bridging the Diagnostic Gap with Convolutional Neural Networks

Research Proposal Presentation

Nepal's Three-Level Healthcare System Faces Critical Gaps

Structural barriers to early cancer detection

Level 1: Primary

Rural health posts and PHCCs. First point of contact for villagers. Staffed by nurses and health assistants.

Level 2: Secondary

District and provincial hospitals. Medical officers and limited specialists. Basic lab tests and surgery.

Level 3: Tertiary

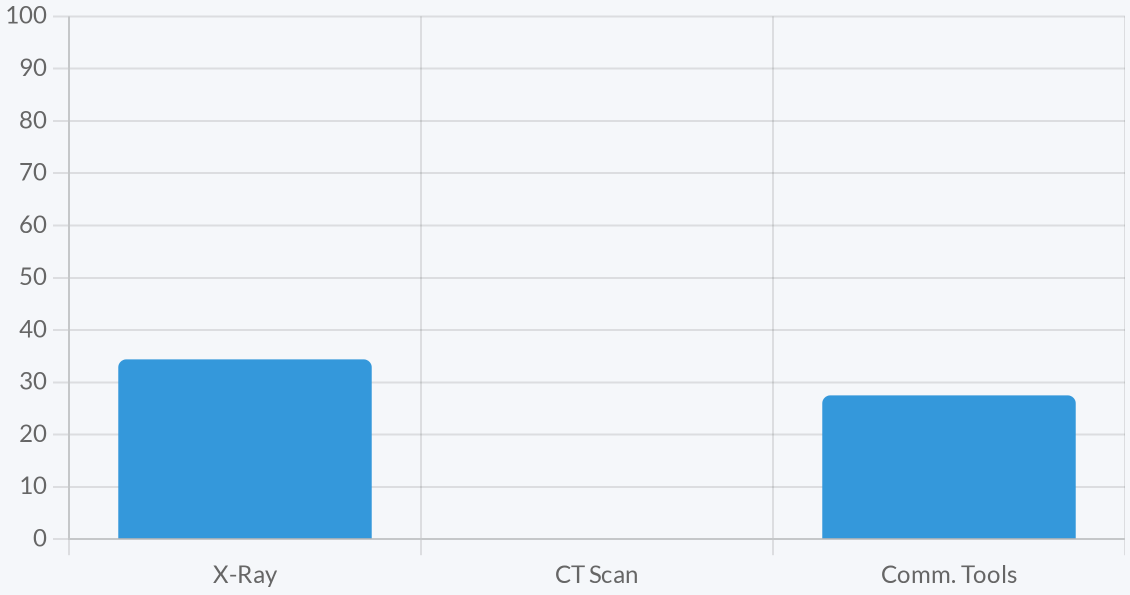
Teaching hospitals and cancer centers in cities. Advanced diagnostics (CT/MRI) and specialized treatments.

⚠ The Core Problem

Patients often present with late-stage cancer because early warning signs are missed at Level 1, where specialized diagnostic expertise is unavailable.

Rural Healthcare Challenges Delay Cancer Diagnosis

Diagnostic Tool Availability in PHCCs (%)



- **Geographic Hurdles:** Patients face 4-5 hour bus rides to reach secondary or tertiary care.
- **Communication Barriers:** Only 25-30% of facilities have reliable links to higher-level hospitals.
- **Economic Strain:** Travel and treatment costs lead to "wait and see" attitudes until it's too late.

Consequence: Small lumps are ignored until they spread, significantly reducing the chances of a complete cure.

Ultrasound Training is Not Enough Without Diagnostic Support

The critical gap between image capture and interpretation

Current Progress

The government has successfully trained rural nurses to perform basic thyroid ultrasounds (USG). However, taking a clear picture is a mechanical skill that differs fundamentally from clinical interpretation.

The Skill Gap:

Most rural health workers lack the specialized training required to identify subtle sonographic signs of malignancy in thyroid nodules.

Systemic Inefficiency

Without diagnostic support, the healthcare system suffers from two major types of errors:

- **Over-referral:** Harmless nodules are sent to city hospitals, causing unnecessary crowding and patient stress.
- **Under-referral:** Real cancer cases are missed or delayed, allowing the disease to progress to advanced stages.

Leveraging the DDTI Dataset for AI Training

High-quality, expert-validated data for model development

Dataset Composition

Source: Digital Database of Thyroid Ultrasound Images (Pedraza et al., 2015).

Volume: 347 real ultrasound pictures from 299 patients.

Diversity: Includes benign (137), malignant (21), and indeterminate cases.

Format: High-resolution scans taken in clinical settings.

Expert Validation

Nodule Mapping: Exact boundaries defined by expert radiologists using XML polygon lines.

TI-RADS Scoring: Standardized assessment of shape, edges, and calcifications.

Ground Truth: All cases confirmed via Fine-Needle Aspiration Biopsy (FNA).

Bethesda System: Standardized cytopathology reporting for final diagnosis.

Dataset Summary Statistics

347	299	100%
TOTAL IMAGES	UNIQUE PATIENTS	BIOPSY CONFIRMED

A Simple Five-Step AI Workflow for Village Health Centers

From image capture to life-saving decisions

STEP 01

Image Capture

Nurse takes a standard thyroid ultrasound picture using existing equipment at the health post.

STEP 02

Preprocessing

Image is automatically resized to 224x224 pixels and brightness is corrected for consistency.

STEP 03

Augmentation

Rotation and flipping help the AI learn from imperfect real-world images taken in rural settings.

STEP 04

AI Analysis

The Convolutional Neural Network (CNN) processes the image to identify malignant features.

STEP 05

Actionable Output

Provides a clear classification ("Safe" vs "Cancer") and a percentage risk score for the nurse.

The entire process is automated, providing immediate feedback to the rural health worker.

Clear Decision Support Empowers Rural Health Workers

Risk-based suggestions for optimized patient referral

Low Risk

Score < 30%

"Everything looks fine. Keep checking every 3–6 months in the village."

Moderate Risk

Score 30% – 60%

"Please send the patient to the district hospital for more tests."

High Risk

Score > 60%

"Send immediately to the cancer hospital – this needs urgent care."

Human-in-the-Loop Approach

The AI acts as a "second pair of eyes" to help nurses make better first decisions. It never replaces the doctor but provides critical support where specialist expertise is absent.

CNN with Transfer Learning Maximizes Accuracy from Small Data

Overcoming medical data scarcity with advanced AI techniques

Phase 1: Pre-training

The computer first learns to recognize basic visual features like **edges, shapes, and textures** from millions of general images (ImageNet). This builds a foundational "visual intelligence."

Phase 2: Fine-tuning

Like a **"smart student"** who already knows general knowledge, the model is then specialized using the specific DDTI thyroid ultrasound dataset to identify malignant nodules.

Performance Expectations

Targeting **85–90% accuracy**, with a primary focus on **high sensitivity**. In medical screening, missing a real cancer case is the most critical error to avoid.

AI Integration Transforms Patient Outcomes in Nepal

Practical solutions for a healthier, more equitable future

Empowered Nurses

Rural health workers gain diagnostic confidence, reducing errors and ensuring that patients receive the right care at the right time.

Economic Relief

Families save significant time and money by avoiding unnecessary, expensive trips to city hospitals for harmless nodules.

System Optimization

Tertiary hospitals can focus on critical cases, reducing wait times and overcrowding while the three levels of care work smoothly together.

Life-Saving Potential

Early detection leads to higher cure rates and longer, healthier lives for thousands of people living in remote areas of Nepal.

"This is not just a student project – it is a practical solution that can be used in real health posts to save lives one day."